

BT5240 - Computational Systems Biology

Assignment 3

Due date: April 10, 2026, 17:00 hrs

Instructions

If you need any assistance with computing, feel free to approach the TAs. Evaluation will be based on the code(s), the answers, and the methodology.

Academic Integrity: Discussing problems verbally with friends is allowed, but copying or looking at codes is not permitted (use of ChatGPT/LLMs is also NOT permitted). Transgressions are easy to find and will be reported and dealt with strictly. Mention any collaboration (discussions only!) in your solutions. Late submission penalties: 1 second to 24 h: 20%; 24 to 48 h: 40%; >48h: 60% Early submission bonuses: >24h: 10%, >48h: 20%

Submission

Join the class on Turnitin using class code 52127280 and enrollment key bt5240_2026. You need to make two submissions: a Python file and a written report.

1. Code guidelines:

- Submit a single .py file containing the code for all the questions.
- Ensure the code is clean and properly formatted.
- Use clear comments to explain logic when necessary.

2. Report guidelines:

- Prepare a report that includes answers to all the questions, all required plots/tables properly labeled, clear explanation of the results.
- Submit a single PDF file. Use Times New Roman, font size 12, 1.5 line spacing, with clean and consistent formatting.

Q1. FBA fundamentals:

While genome-scale models are often used to predict growth, cellular metabolism rarely operates with growth as its only objective. In many biological settings, carbon flux must be shared between biomass production and the synthesis or secretion of other metabolites. Changing this balance can alter pathway usage, reveal hidden bottlenecks, and expose vulnerabilities that are not obvious under standard growth optimisation.

1. Download the iJO1366 model (*Escherichia coli*) from BiGG Models and load it. Find the minimal media of the model and use it to simulate the following scenarios. Under both aerobic and anaerobic conditions, perform:

- i. flux balance analysis (FBA)
- ii. parsimonious flux balance analysis (pFBA)
- iii. flux variability analysis (FVA)

From your analysis:

- a. Report the biomass obtained under aerobic and anaerobic conditions.
 - b. Compare whether FBA and pFBA produce identical or different flux distributions.
 - c. Identify the reactions that retain wide allowable flux ranges in FVA despite unchanged biomass.
 - d. Briefly comment on the distinct insights provided by FBA, pFBA, and FVA, and the contexts in which each method is most useful.
2. The biomass-maximising solution does not necessarily favour metabolite secretion. Modify the model to promote ethanol production alongside biomass formation - construct a combined objective function using the original biomass reaction together with the ethanol secretion reaction (for example, by assigning different weights to each term). You may test any weighting scheme, but report results for at least two different weight combinations. Under both aerobic and anaerobic conditions:
 - a. Describe your modified objective function and report the final biomass and ethanol secretion flux.
 - b. Compare how strongly biomass changes relative to the original growth-maximising solution and Identify reactions whose flux changes substantially when ethanol production is introduced.
 - c. Briefly comment on how oxygen availability influences the balance between growth and ethanol production.

Q2. Cross-pathogen target discovery:

Bacterial pathogens continue to remain a major challenge because of their ability to adapt quickly to changing environments and therapeutic interventions. Some survive for long periods inside host tissues under highly restricted conditions, while others thrive in clinical settings by exploiting metabolic flexibility and resistance mechanisms. Despite these differences, every pathogen faces the same basic requirement: it must use whatever nutrients are available to sustain growth, maintain cellular functions, and reproduce. Studying metabolism at the network level allows us to ask whether very different

organisms rely on common biochemical strategies, and whether these shared requirements can reveal useful intervention points.

Mycobacterium tuberculosis is a slow-growing pathogen adapted to long-term survival within the host, often under nutrient and oxygen-limited conditions. In contrast, ***Acinetobacter baumannii*** is an opportunistic hospital-associated pathogen known for rapid adaptation and multidrug resistance. Their lifestyles are clearly different, but both must maintain their core metabolic processes that support growth.

1. Download the models iNJ661 (*Mycobacterium tuberculosis*) and iCN718 (*Acinetobacter baumannii*) from BiGG Models (use a format compatible with your system; '.xml' is recommended). Compare the two models in terms of genes, metabolites, and reactions, and examine whether differences in gene content are reflected proportionally in overall network size. In particular:
 - a. Does a higher number of genes correspond to a higher number of reactions and metabolites? Which organism shows a larger reaction-to-gene ratio?
 - b. Briefly comment on how the lifestyle of each organism may explain the differences or similarities you observe.
2. Identify essential genes in both models by testing the effect of individual gene deletions on growth. From your results:
 - a. List the essential genes identified in each organism.
 - b. Identify the essential genes common to both models (with matching gene annotations). Examine whether these shared genes are associated with central metabolism, biosynthetic pathways, transport functions, or other cellular processes.
3. The genes found to be essential in both organisms represent potential common targets for metabolic intervention. Choose one shared essential gene and examine it in more detail:
 - a. Identify the metabolic function associated with the gene and briefly mention why disrupting this function is expected to strongly affect growth in both organisms.
 - b. Briefly examine available literature to see whether inhibition of this gene, or the pathway it belongs to, has been associated with reduced growth or loss of viability in either organism or in related bacterial species.

Q3. Synthetic lethality across diets:

Metabolic networks often contain hidden redundancies that are not apparent under single gene deletions. In many cases, loss of a single gene can be compensated for by an alternative pathway, while simultaneous disruption of two genes removes this backup capacity and abolishes growth.

1. Download the [Escherichia coli core](#) model from BiGG Models, load it and perform basic validation checks expected for a genome-scale metabolic model before beginning any simulations. As part of model validation:
 - a. Report the number of genes, metabolites, and reactions before and after validation.
 - b. Describe the checks performed and any corrections made to the model.
 - c. Employ the algorithm taught in class to identify the gene knockout targets - (single genes, pairs and triplets) that would enhance the production of ethanol.
2. Identify synthetic lethal gene pairs by implementing the FAST-SL algorithm (use the [fastsl python package](#)). Perform the analysis under the following growth conditions:
 - i. Glucose as the sole carbon source ($10 \text{ mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$)
 - ii. Acetate as the sole carbon source ($10 \text{ mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$)
 - iii. In a medium comprising only the components absolutely necessary for growth

From your results:

- a. List the synthetic lethal gene pairs identified under each condition.
 - b. Identify gene pairs common to all three media, as well as those specific to individual conditions.
 - c. Briefly comment on whether synthetic lethality depends on the media condition, i.e., the nutrient environment.
3. Select one synthetic lethal gene pair and explain:
 - a. Why does the deletion of either of the genes still allow growth and why does the simultaneous deletion inhibit biomass production?
 - b. Which redundant metabolic route or alternative pathway is disrupted in the double deletion?